Qiancheng (Sam) Sun

www.linkedin.com/in/qiancheng-sun | samuelsun2003@gmail.com | +1 (662) 497-0904 | https://samsunportfolio.wixsite.com/my-site-1

EDUCATION

Purdue University | West Lafayette, IN *2022-2027*
Bachelor of Science, Mechanical Engineering GPA: 3.89 / 4.00

EXPERIENCE

Formlabs | Somerville, MA *Jun 2025 – Aug 2025*
Manufacturing Design Engineering Intern

- Owned the full **electromechanical** design of a scalable automated solution that accurately measures linear rail drag force and straightness for all current and future SLA printers with a repeatability of **0.3N** and **5 microns**. Reduced rework frequency by **~20%**
- Performed hands-on prototype assembly and testing for a new large-format **SLS** product line; worked closely with design teams to implement **DFA** improvements, reducing assembly time by ~33% on targeted subassemblies
- Conducted **Gauge R&R** on a planarity jig and used **CMM** to root cause errors. Analyzed results in **JMP** to validate jig performance.

Tesla | Austin, TX *Aug 2024 – May 2025*
Mechanical Design & Manufacturing Engineering Intern

- Owned end-to-end design and implementation of various production devices, from prototyping to deployment, to address safety risks and improve efficiency. Created detailed **MIs** and **ECOs** to support adoption and sustainment. Designs include:
  - Dissector device that reduces quality dissection time by ~50% while also improving accuracy and consistency
  - Blower and Exhaust system that reduces containments by 74% and cleaning frequency by 62% in the laser welding process
  - Camera brace and Micrometer guard that utilizes compliant mechanisms to reduce costs by 15% and install time by 80%
- Led the standardization & rollout of a precision air jet system, replacing flawed hardware for a critical step of **4680** winding process
  - Performed **DOE** to identify and eliminate unnecessary components on gimbal; reduce wear and increased the process window by 100+%; conducted **tolerance stack analysis** to ensure consistent performance across manufacturing variation
  - Spearheaded first-principles investigation into hi-pot defects caused by Gen 2 air jets; used **CFD** and high-speed footage to isolate root cause and deploy solution, restoring **~1% yield** fleet-wide.
  - Identified and redesigned critical components to resolve 12+ **DFA**-related failures and prevent recurrence.
- Collaborated with **OEM** to design custom one-way rollers with fiber optic sensors for missed roller detection, reducing scrap by 10,000+ cells/month; iterated designs through supplier **DFM** reviews to deliver a low-cost, manufacturable solution.
- Built custom **SQL** queries & **Tableau** dashboards to monitor downstream impact of design changes, supporting data-driven decisions

Panasonic Energy of North America (PENA) | Sparks, NV M*ay 2024 – Aug 2024*
Equipment Engineering Intern

- Designed a 3-axis adjustable mounting bracket using **DFMA** principles for inline vision inspection system testing on high-speed winders for 2170 battery cells for Tesla Model Y.
  - Sourced and procured parts from vendors using detailed engineering drawings with **GD&T** and welding callouts, delivering a design that was 32% under budget
- Collaborated with supplier to select and integrate a sensor for metallic contaminant detection; used FMEA to mitigate risks. Initiated automation of manual inspection, projecting 24+ hour time savings and prevention of 300,000+ defective cells annually.

Semiconductor Packaging Laboratory | *Cooling Structure Stress Analysis Researcher* *Apr 2024 – May 2024*

- Used Representative Volume Element method (RVE) in **COMSOL** to model and analyze the mechanical properties and stress region of custom copper cooling structure. Optimized CAD structure to maximize reliability and cooling effectiveness

Projects

Accessible 3D Printed Prosthetic Hand | *Formlabs Hackathon* *2022*

- Designed a <$200 prosthetic hand using single servo and 3D printed gears reduce cost and simplify assembly for broader accessibility
- Engineered compliant fingers with flexible resin and elastic couplings to enable adaptive gripping behavior while minimizing gear stress, despite limited individual articulation.
- Developed a modular arm interface with integrated electronics housing and adjustable fittings to accommodate diverse limb differences; included a demo adapter for use by able-bodied users during outreach and testing.

AI Powered Smart Trash-Can | *MakeMIT 2023 | MIT Hardware Hackathon* *2023*

- Built and presented prototype trash can within 24 hours; Second place MakeMIT award, 1517 Medici Grant Recipient
- Auto-opening lid, Computer Vision powered object recognition, Automatic Trash sorting

SKILLS

CAE: SolidWorks, OnShape, Fusion 360, NX, COMSOL, FEA, GD&T, ANSYS, CFD | **Fluent Languages:** Mandarin & English
Mechanical: Design for Manufacturing/Assembly (CNC, Sheet Metal, 3D Printing), PDM, Root Cause Analysis, FMEA, Tolerance Stack
Software/Electrical: Python, Matlab, C++, Linux, Arduino, Raspberry Pi, Soldering, SQL, JMP, Tableau, JIRA, Confluence